

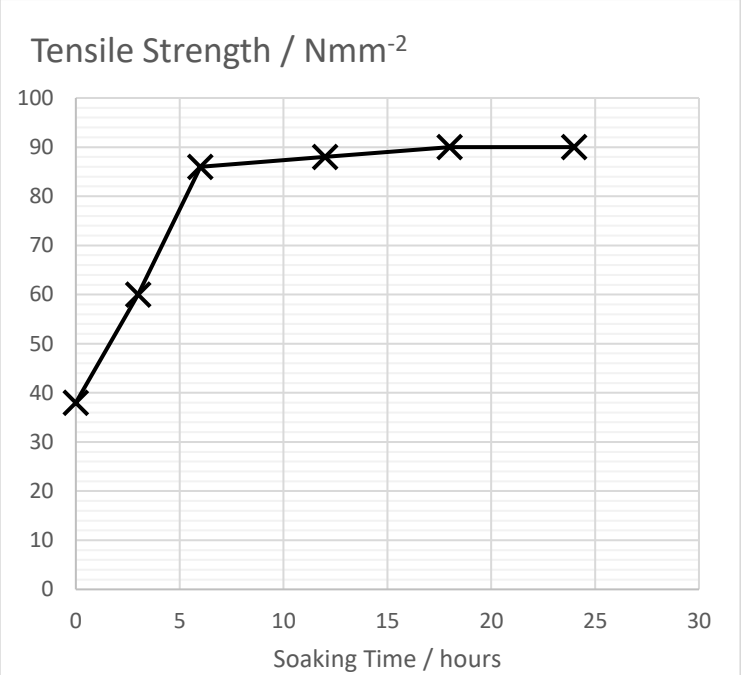
Question Number	Answer	Additional Guidance	Mark
1ai	An answer that includes the following • sclerenchyma (1) • xylem (vessel) (1)	IGNORE phloem, companion cells, cambium, parenchyma	grad 2

Question Number	Answer	Additional Guidance	Mark
1aii	An answer that includes the following: • (increased) {strength / support / hardness / rigidity} (1) • makes cell (wall) impermeable / reduces permeability or description e.g., stops / reduces water entering or leaving (1)	IGNORE flexible ACCEPT stops anything going in or out / makes (cell wall) waterproof	expert 2

Question Number	Answer	Additional Guidance	Mark
1bi	• (soaking) time (1)		grad 1

Question Number	Answer	Additional Guidance	Mark
1bii	An answer that includes the following points: • cut {thin / transverse} section (of fibre) / lay (fibre on slide) on microscope (1) • measure using (eyepiece) graticule (1) • which has been calibrated (with stage micrometer) (1)	ACCEPT description of calibration	expert 3

Question Number	Answer	Additional Guidance	Mark
1biii	An answer that includes the following points • fibre {diameter / radius} is needed to calculate (cross sectional) area (1) • (cross sectional) area is needed to calculate tensile strength / strength of fibre (1) • diameter will affect {tensile strength / strength of fibres} (1)	ACCEPT diameter is needed to calculate tensile strength for 1 mark	expert 2

Question Number	Answer	Additional Guidance	Mark
1ci	A graph with the following features • A axes correct (x - soaking time, y- tensile strength) (1) • L all labels correct (x- soaking time / hours, y- tensile strength / Nmm^{-2}) (1) • P plots correct on a linear scale on both axes (1) • S points joined with straight lines (1)	They lose mp S if they extrapolate beyond 24 hours. They can split the y-axis as long as they show it clearly, or they can start at numbers above zero Neither axis has to start at zero, just make sure it is linear 	expert 4

Question Number	Answer	Additional Guidance	Mark
1cii	An description that includes three of the following points - soaking time increases (tensile) strength (1) - relationship is (nearly) linear {up to 6 hours / from 6 to 18 hours} (1) {large / significant / rapid} increase up to 6 hours / more than doubles up to 6 hours (1) - slow increase (after 6) / no increase after 18 hours / levels off (1)	DO NOT piece together, must be a clear statement ACCEPT manipulation to show size of increase	expert 3